Integral Source Completion and a Finite Local–Global Decision Principle for Stationary CCR Polynomial Jets

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Abstract

For the formal stationary chronological-carry refinement (CCR) polynomial transport, we give an exact criterion for simultaneous vanishing of its first three coefficients with ordinary integer sources. The last three sources admit a unique explicit rational completion. The upper half of all source coordinates enters the three-jet affinely, yielding a rank-three integer-lattice membership criterion after arbitrary fixed lower sources. A rational-polynomial terminal map then gives an explicit finite common denominator period and a local–global equivalence for fixed order and transport prime: an ordinary-integer completion exists precisely when local integral completions exist at a finite set of primes. For odd order n and a Bernoulli-admissible transport prime $P > 2n$, only auxiliary primes at most n can obstruct. We exhibit exact source-dependent denominator exponents for order 77 as sharp coefficient-denominator instances of the general completion algorithm. This local–global principle relies on the special finite-denominator source parametrization, and should not be confused with a general Hasse principle for Diophantine varieties.

Keywords: CCR transport; integer-source completion; p-adic lifting; Bernoulli numerator; Smith normal form; rational-polynomial integrality; effective local–global principle.

Contributions and novelty

The main contributions concern the explicit *formal stationary CCR source family*, not new general forms of Smith normal form, polynomial denominator clearing, or the Chinese remainder theorem. First, an exact terminal-source triangularity theorem produces the unique rational three-source completion and explicit integrality conditions for the bottom three polynomial coefficients. Second, source-weight filtration shows that the upper half of the sources acts affinely, giving a concrete integral image-lattice criterion. Third, coefficientwise denominator control turns the unrestricted integer-source completion question, at each fixed order and transport prime, into a finite local–global decision problem; under the specified Bernoulli-admissibility condition only auxiliary primes at most the order are needed. Fourth, the order-77 calculations give exact sharp coefficient-denominator exponents for the two Bernoulli-admissible transport primes, without claiming an actual global YES/NO decision there.

The CCR Sheffer representation and the elementary lattice and CRT principles are acknowledged as prior structural tools. The novelty claimed here lies in the simultaneous three-jet completion map and its application to this precisely stated integral-source problem; no all-order triple-zero impossibility or physical realization theorem is asserted.

1 Problem and precise scope

For a parameter P and source variables λ_s, let $F_0(X) = 1$, $F_j = 0$ for $j < 0$, and define $F_j \in \mathbb{Q}[P, \boldsymbol{\lambda}, X]$ by

$$\begin{aligned} T_jF_j(X) &= ((P+1)X - P)F_{j-1}(X) + P((P+1)(X-1) - P)F_{j-2}(X-1) \\ &\quad + P\sum_{s=3}^{j}(-1)^s\lambda_s F_{j-s}(X-s+1), \\ T_jG(X) &= XG(X+1) - (X-j)G(X). \end{aligned} \tag{1}$$

As $T_j(X^k) = (j+k)X^k + \text{lower-degree terms}$, the transport has a unique solution over $\mathbb{Q}$. Fix odd $n \geq 5$ and a prime $P > 2n$. Our problem is

$$\exists\,(\lambda_3, \ldots, \lambda_n) \in \mathbb{Z}^{n-2} : \quad [X^0]F_n = [X^1]F_n = [X^2]F_n = 0. \tag{2}$$

We distinguish ordinary-integer, local integral, and rational local sources. The results below give an exact finite decision reduction for each fixed (n, P) without asserting a uniform bound on complexity across orders.

2 CCR response series and high-source filtration

Write $G(t, X) = \sum_{j\geq 0} F_j(X)t^j$. Comparing the X-linear and constant terms after summing the recurrence gives a uniquely normalized Sheffer representation

$$G(t, X) = H_0(t)\exp(Xf(t))\exp\left(\sum_{s\geq 3}\lambda_s K_s(t)\right), \qquad f = \log Y, \quad H_0(0) = Y(0) = 1. \tag{3}$$

The source-response kernel is the exact rational series

$$K_s(t) = P(-1)^s\int_0^t z^{s-1}Y(z)^{-(s-1)}\,dz = \frac{P(-1)^s}{s}t^s + O(t^{s+1}). \tag{4}$$

With $a = (P+1)/2$ and $D = (P+1)(5P+1)/4$, the source-free series is

$$Y(t) = 1 + at + \sum_{k\geq 1}\frac{2^{2k}B_{2k}D^k}{(2k)!}\,t^{2k}. \tag{5}$$

This is the stationary CCR deformation: the mere Sheffer form is classical, not new. In particular

$$[X^j]F_n = [t^n]H_0(t)\frac{f(t)^j}{j!}\exp\left(\sum_{s=3}^{n}\lambda_s K_s(t)\right). \tag{6}$$

Since $\operatorname{ord}_t K_s = s$ and $\operatorname{ord}_t f = 1$, any source monomial $\prod_s \lambda_s^{e_s}$ occurring in the j-th coefficient necessarily satisfies

$$\sum_{s\geq 3} se_s \leq n - j. \tag{7}$$

This elementary filtration is the key extra structural input.

Theorem 2.1 (Terminal jet triangularity)**.** *For $1 \leq r \leq n-2$, put $d_{n,j} = a^{-j}[X^j]F_n$ for $0 \leq j < r$. With the first $n-r-2$ source coordinates fixed, the map from $(\lambda_{n-r+1}, \ldots, \lambda_n)$ to $(d_{n,r-1}, \ldots, d_{n,0})$ is affine triangular with constant diagonal entries*

$$\gamma_{n,j} = \frac{P(-1)^{n-j}}{(n-j)j!}, \quad j = r-1, \ldots, 0.$$

Consequently its Jacobian determinant is

$$\det M_{n,r} = \frac{(-1)^{rn-r(r-1)/2} P^r}{\prod_{j=0}^{r-1}(n-j)j!} \neq 0.$$

For $r = 3$ and odd n this specializes to $P^3/(2n(n-1)(n-2))$.

Proof. Since $f(t) = at + O(t^2)$, the leading term of the λ_{n-j} contribution to (6) is obtained uniquely from $t^{n-j} \cdot (at)^j/j!$. By (4) its coefficient is $P(-1)^{n-j}a^j/((n-j)j!)$. Sources of index greater than $n-j$ do not occur. Multiplication of the triangular diagonal entries gives the determinant. The bound $P > 2n$ ensures $a \neq 0$ and permits the normalization. □

Corollary 2.2 (Unique rational completion)**.** *Fix any earlier sources $u = (\lambda_3, \ldots, \lambda_{n-3}) \in \mathbb{Q}^{n-5}$. There exists a unique triple $W(u) \in \mathbb{Q}^3$ for the final sources $(\lambda_{n-2}, \lambda_{n-1}, \lambda_n)$ that makes all three lowest jets zero. Each coordinate of W is a rational-coefficient polynomial in u.*

Proof. By Theorem 2.1, the final three columns form a constant invertible rational triangular matrix and the remaining jets are rational polynomials in u. Solving that matrix equation proves both assertions. □

3 Explicit three-source integrality formulas

Let $s = n-2$, $t = n-1$, and set the last three sources to zero temporarily. Write $b_j = a^{-j}[X^j]F_n\big|_{\lambda_s=\lambda_t=\lambda_n=0}$, for $j = 0, 1, 2$. Set $\mu_k = (-1)^k\lambda_k$ for the three terminal sources and define the integer polynomials

$$\begin{aligned} A &= 6Ps^2 - Ps + 5P + 6s^2 - 5s + 1, \\ C &= Ps^2 + 3Ps + 4P + s^2 + s, \\ D_0 &= 3P^2s^3 + 12P^2s^2 + 17P^2s + 44P^2 \\ &\quad + 6Ps^3 + 12Ps^2 + 8Ps + 28P + 3s^3 - s. \end{aligned}$$

(The term $8Ps$ is important; changing it to $8P$ is incorrect.)

Theorem 3.1 (Pivot-free integrality criterion)**.** *The unique terminal signed sources satisfying the three-jet equation are*

$$\boxed{\begin{aligned} \mu_s &= -\frac{2s}{P}b_2, \\ \mu_t &= -\frac{t}{P}b_1 - \frac{A}{6P}b_2, \\ \mu_n &= -\frac{n}{P}b_0 - \frac{C}{2P}b_1 - \frac{D_0}{12P}b_2. \end{aligned}} \tag{8}$$

For the fixed early vector, ordinary-integer completion exists if and only if all three displayed rational values are integers. Equivalently, for every prime q the three numerator forms have the valuations needed to cancel their denominators P, $6P$ and $12P$, respectively.

Proof. For clarity, the *entire* constant terminal response matrix can be written explicitly. Put

$$\begin{aligned} B = (P+1)(&3Ps^4 + 2Ps^3 - 3Ps^2 - 50Ps - 24P \\ &+ 3s^4 - 2s^3 - 3s^2 + 2s). \end{aligned}$$

In decreasing jet order, and with $t = s + 1$ and $n = s + 2$, the exact normalized response is

$$\begin{pmatrix} d_{n,2} \\ d_{n,1} \\ d_{n,0} \end{pmatrix} = \begin{pmatrix} b_2 \\ b_1 \\ b_0 \end{pmatrix} + P \begin{pmatrix} \frac{1}{2s} & 0 & 0 \\ -\frac{A}{12st} & \frac{1}{t} & 0 \\ \frac{B}{24stn} & -\frac{C}{2tn} & \frac{1}{n} \end{pmatrix} \begin{pmatrix} \mu_s \\ \mu_t \\ \mu_n \end{pmatrix}.$$

The first row gives $\mu_s = -2sb_2/P$; substitution into the second gives $\mu_t = -tb_1/P - Ab_2/(6P)$. In the third row the remaining b_2 coefficient is $(B - AC)/(12tP)$. Direct expansion of the displayed integer polynomials gives the exact identity

$$B - AC = -tD_0 \qquad \text{in } \mathbb{Z}[P, s],$$

which yields the last expression in (8). The triangular matrix has nonzero diagonal, so the rational completion is unique. Because changing each λ_k to $\mu_k = (-1)^k \lambda_k$ preserves integrality, an integer completion exists precisely when all three rational expressions are integers. □

4 Half-source affine elimination and a lattice criterion

Put $h = \lfloor n/2 \rfloor$ and split

$$u = (\lambda_3, \ldots, \lambda_h), \qquad v = (\lambda_{h+1}, \ldots, \lambda_n).$$

Theorem 4.1 (Source-half elimination)**.** *For $j = 0, 1, 2$ there are rational polynomials $b_j(u)$ and $A_{js}(u)$ such that*

$$[X^j]F_n = b_j(u) + \sum_{s=h+1}^{n} A_{js}(u)\lambda_s. \tag{9}$$

Moreover A_{js} only depends on low sources of indices $i \le n - s - j$, and is zero for $s > n - j$.

Proof. Two occurrences of a high source (possibly with different indices) have t-order at least $2(h + 1) > n$ and cannot contribute to the coefficient of t^n. The support refinement is the degree budget (7). Hence the dependence is affine in all the high-source variables simultaneously, including mixed terms with arbitrary low sources. □

For a fixed *integer* low-source vector u, select positive row multipliers D_j that clear the rational denominators of the coefficients of every b_j and A_{js} as polynomials in u. Let $M(u) \in \mathbb{Z}^{3\times(n-h)}$ and $B(u) \in \mathbb{Z}^3$ denote the resulting integer matrix and constant vector. The terminal three-source minor is invertible by Theorem 2.1, so $M(u)$ has rank three.

Theorem 4.2 (Determinantal integer completion)**.** *Write $g(M)$ for the positive gcd of all 3×3 minors of a full-row-rank integer matrix. Then, for every fixed $u \in \mathbb{Z}^{h-2}$,*

$$\exists v \in \mathbb{Z}^{n-h} \text{ with } B(u) + M(u)v = 0 \quad \Longleftrightarrow \quad \boxed{g(M(u)) = g([M(u) \mid B(u)])}.$$

Proof. The image $L = M\mathbb{Z}^{n-h} \subseteq \mathbb{Z}^3$ has index $g(M)$, by Smith normal form. The enlarged lattice $L' = L + B\mathbb{Z}$ has index $g([M \mid B])$. Since $L \subseteq L'$, the indices are equal if and only if $L = L'$, equivalently $B \in L$. The sign is immaterial. □

Remark 4.3. *Smith normal form and lattice-index equalities are classical. The CCR-specific content is the exact filtration guaranteeing a* constant-rank high-source affine matrix *at every order;* (9) *and the terminal pivot give a finite-dimensional concrete instance. We make no general novelty claim for the lattice theorem absent CCR structure.*

5 Effective finite local–global principle

Write the signed or unsigned terminal triple in Corollary 2.2 as $W = (W_1, W_2, W_3)$, polynomials in the $n-5$ early sources u. Define

$$E^*(n,P) = \operatorname{lcm}\{\operatorname{den}(w_{i,\alpha}) : W_i(u) = \sum_{\alpha} w_{i,\alpha} u^{\alpha},\ 1 \le i \le 3\}. \tag{10}$$

The denominators are positive integers in reduced terms. Notice that E^* is *the least positive integer clearing every rational polynomial coefficient*; we do not assert that it is the least period of the Boolean integer-solvability predicate.

Lemma 5.1 (Exact completion periodicity)**.** *For every $u, u' \in \mathbb{Z}^{n-5}$ with $u \equiv u' \pmod{E^*}$ coordinatewise,*

$$W(u') - W(u) \in \mathbb{Z}^3, \quad \textit{hence} \quad W(u') \in \mathbb{Z}^3 \iff W(u) \in \mathbb{Z}^3.$$

Proof. Each $E^* W_i$ has integral polynomial coefficients. Therefore its values at congruent inputs are congruent modulo E^*, and division by E^* gives the integer difference. □

Theorem 5.2 (Finite local–global decision for fixed n, P)**.** *The following assertions are equivalent:*

1. *Some ordinary integer source vector has $X^3 \mid F_n$ over $\mathbb{Q}[X]$.*
2. *For every prime q, some q-adic integral source vector has $X^3 \mid F_n$ over $\mathbb{Q}_q[X]$.*
3. *For each prime $q \mid E^*$, there exists an early source vector $u_q \in \mathbb{Z}_q^{n-5}$ with $W(u_q) \in \mathbb{Z}_q^3$.*
4. *There exists an early residue tuple $u \in (\mathbb{Z}/E^*\mathbb{Z})^{n-5}$ with $W(u) \in \mathbb{Z}^3$.*

For fixed (n, P), the test in (4) is finite, although its cost may be enormous.

Proof. (1)⇒(2) is immediate, and (2)⇒(3) uses the uniqueness of the terminal tuple in $\mathbb{Q}_q$. For (3)⇒(4), factor $E^* = \prod_{q \mid E^*} q^{e_q}$ and, for each $q \mid E^*$, approximate u_q by integer residues modulo q^{e_q}. Combine those residues coordinatewise by CRT to get an integer u. Since $q^{e_q} W_i \in \mathbb{Z}_q[u]$, the difference $W_i(u) - W_i(u_q)$ lies in $\mathbb{Z}_q$. Thus $W_i(u)$ is q-adically integral at all primes dividing E^*; at primes not dividing E^*, all coefficients of W_i are already integral. A rational number integral at every rational prime is an integer, so $W(u) \in \mathbb{Z}^3$. Taking the class modulo E^* gives (4). Finally (4)⇒(1) by substituting the unique terminal triple. The periodicity lemma makes the residue formulation independent of the chosen representatives. □

Remark 5.3. *The theorem is an effective restricted local–global principle for this* polynomially parametrized, terminally invertible *problem. It is a consequence of denominator clearing and CRT, not a general local–global theorem for arbitrary polynomial equations. Existing general local–global results about polynomial maps must be compared before claiming priority for the abstract mechanism.*

Theorem 5.4 (Only small auxiliary primes can obstruct)**.** *Assume n odd and $P > 2n$ prime, and that the Bernoulli-admissibility condition*

$$P \mid \operatorname{num}(B_{n-1})$$

holds. Then $q \nmid E^*(n,P)$ for every rational prime $q > n$. In particular an ordinary-integer triple-zero completion exists if and only if *local $\mathbb{Z}_q$-integral completions exist for each prime $q \le n$.*

Proof. Fix a prime $q > n$ with $q \neq P$. We work with series truncated through degree n over $\mathbb{Z}_q$. Division by each integer $1, \dots, n$ is allowed. The denominators of the Bernoulli terms occurring in (5) are q-adic units: by von Staudt–Clausen, a prime q in the denominator of B_{2k} would require $q - 1 \mid 2k$; but $2k \leq n$ and, since $n \geq 5$ is odd, every prime $q > n$ has $q - 1 > n$. Set $a = (P + 1)/2$; the identity $D = a(5P + 1)/2$ therefore implies

$$Y(t) - 1 \in a\, t\mathbb{Z}_q[t]/(t^{n+1}), \qquad f(t) = \log Y(t) \in a\, t\mathbb{Z}_q[t]/(t^{n+1}).$$

Even when $q \mid (P + 1)$, the quotient f/a consequently has q-integral coefficients and leading term t. The source-free relation

$$(\log H_0)' = -P - P(2P + 1)t/Y$$

and the kernel formula (4) show that H_0 and each K_s, truncated through degree n, have q-integral coefficients; exponentials preserve this property to degree n because $1, \dots, n$ are units. Hence the *normalized* jets are coefficientwise q-integral polynomials in the sources:

$$d_{n,j} = [t^n]H_0(t)\frac{(f(t)/a)^j}{j!}\exp\left(\sum_{s=3}^{n}\lambda_s K_s(t)\right) \in \mathbb{Z}_q[\lambda_3, \dots, \lambda_n] \quad (j = 0, 1, 2).$$

The constant triangular terminal pivots $\gamma_{n,j} = P(-1)^{n-j}/((n - j)j!)$ are q-adic units for $q > n$, $q \neq P$; recursive elimination therefore gives $W \in \mathbb{Z}_q[u]^3$. In particular, no division by the potentially nonunit a is needed after normalization.

At $q = P$, the reduction of the CCR recurrence modulo P is source-independent and equals the classical Nörlund–Stirling shadow. Its first two jets vanish at odd order, and the third vanishes by the assumed $P \mid \mathrm{num}(B_{n-1})$. Thus the low-jet source polynomials have all coefficients divisible by P over $\mathbb{Z}_P$. After factoring P from the terminal response matrix, the remaining triangular diagonal entries are P-adic units (since $P > 2n$), so $W \in \mathbb{Z}_P[u]^3$. Therefore no prime above n divides a coefficient denominator of W. The local–global equivalence follows from Theorem 5.2. □

Remark 5.5. *The Bernoulli condition does not say that an* integer *completion exists. It says that the transport prime itself causes no local obstruction; the small auxiliary primes still need to be tested. The equivalence is fixed-parameter and does not assert a uniform conclusion over unbounded orders.*

6 Exact examples and decision precision

n	P	Role of example	E from RN177	E^*
5	11	Formal nonadmissible test	5940	1485
7	17	Formal nonadmissible test	1156680	385560
9	23	Formal nonadmissible test	10432800	1304100
13	691	Actual Bernoulli-admissible candidate	563415288259824000	6810804000

These coefficientwise periods are computed with exact rational polynomial arithmetic and independently checked against the original transport recurrence. In the admissible $(n, P) = (13, 691)$ case,

$$E^* = 2^5 \cdot 3^5 \cdot 5^3 \cdot 7^2 \cdot 11 \cdot 13,$$

so no prime beyond $n = 13$ can obstruct and the transport prime 691 does not occur. These figures are *not* claims about minimal periods of integer-valued Boolean predicates.

For the explicit higher-order case $n = 77$, Bernoulli admissibility leaves

$$P_1 = 58231, \qquad P_2 = 22284285930116236430122855560372707885169924709.$$

The exact coefficientwise 2- and 3-adic denominator exponents of the three unique signed terminal polynomials are:

P	$(e_2(W_1), e_2(W_2), e_2(W_3))$	$(e_3(W_1), e_3(W_2), e_3(W_3))$
P_1	$(52, 52, 52)$	$(34, 34, 35)$
P_2	$(54, 52, 55)$	$(34, 34, 35)$

These are exact *coefficient-denominator* valuations, proved by complete min-plus bounds over all 72 early sources and sharp rational coefficient witnesses, rather than statements about the least root-lifting depth. They give sufficient finite test moduli 2^{52}, 3^{35} at P_1 and 2^{55}, 3^{35} at P_2. The high-index sources cannot simply be discarded: already λ_{74} can alter the relevant local integrality. The upper half $\lambda_{39}, \ldots, \lambda_{74}$ can instead be handled by the exact affine lattice test.

Additional archived computations give restricted local source certificates at order 77, but they are not used to establish the global theorems of this article.

7 Relation to prior papers and novelty boundary

Earlier CCR-3, CCR-4 and CCR-k studies [1, 2, 3] introduce the chronological-carry setting. The existing articles *CCR-k Arithmetic Refinement: p-Adic Resonance, Ramification, and Arithmetic Density Laws* and *Sharp p-Adic Rigidity and Source-Tunable Resonance in Stationary CCR Polynomial Transports* [4, 5] study, respectively, first-order Stirling-source obstruction lattices and near-leading coefficients, with root-disk deformations and resonance bounds. The present work concerns simultaneous vanishing of the *three lowest coefficients* and ordinary-integer completion of *all* sources. It does not reuse their high-jet resonance theorem as a new contribution, and it cites the previously established Sheffer decomposition instead of claiming its discovery.

The finite-denominator local–global mechanism, Smith index theorem, CRT, and the general theory of rational-polynomial integral values are standard tools. The cited Nörlund denominator results and general CRT/Smith principles remain prior art; the distinctions from the CCR-specific new results are summarized in the opening Contributions and novelty section.

8 Exact computational verification

The symbolic theorems above use explicit triangular elimination, source-weight filtration, integer-lattice indices, and coefficientwise denominator clearing. Independent exact-rational consistency checks previously recorded for the formal transport include 24 terminal matrices (with 216 checked entries), 126 multiplicity tests, and 46 direct terminal-completion substitutions in the original stationary CCR recurrence. Four fixed-order examples, including $(n, P) = (13, 691)$, were checked by coefficientwise rational-polynomial denominator reconstruction and direct recurrence substitution. Such finite example checks corroborate the derivations; they do not establish the universal theorems by sampling.

The order-77 exponent table has a different proof status: it uses finite but exhaustive *coefficient-level* arithmetic for all 72 early-source coordinates. Each source monomial is included in a min-plus valuation bound obtained from exact truncated series; coefficients attaining the reported 2- and 3-adic exponents certify sharpness. In the recorded exact certificate, the 2-adic exponents are attained by constant coefficients, while the 3-adic bounds have constant or λ_3 coefficient witnesses. This is a computer-assisted finite certificate for those exact exponent vectors, not a random sample or a solution of the remaining local existence tests at order 77. The complete high-dimensional coefficient calculation has not been independently rerun in this editorial revision; its archived mathematical outcomes, not an additional external verification claim, are reported here.

No verification code, executable files, or supplementary data archive is included or promised as part of this publication. The mathematical claims and the stated finite computation outcomes, including their respective roles and limitations, are presented in the article.

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Data availability. This is a mathematical study and uses no empirical or personal dataset; the relevant definitions, theorems and symbolic proofs are contained in the article. The finite-arithmetic certificate outcomes underlying the order-77 denominator examples are stated with their scope and limitations in the verification discussion. No supplementary code or data archive is part of this publication.
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